

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a proven allergy to cariprazine

Special Populations

Renal Impairment

- Mild to moderate impairment (creatinine clearance <30 mL/minute):
no dose adjustment necessary
- Severe or end-stage: not recommended

Hepatic Impairment

- Mild to moderate impairment (Child-Pugh score between 5 and 9):
no dose adjustment necessary
- Severe: not recommended

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some elderly patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no atypical antipsychotic has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic

Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using cariprazine may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR

or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- In rats, administration of cariprazine during organogenesis caused malformations, lower pup survival, and developmental delays at exposures less than the human exposure at maximum recommended human dose (6 mg/day); cariprazine was not teratogenic in rabbits at doses up to 4.6 times the maximum recommended human dose
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if cariprazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk

- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on cariprazine should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- For patients who do not tolerate aripiprazole or brexpiprazole for schizophrenia
- For patients who do not tolerate quetiapine, lurasidone, or olanzapine for bipolar depression
- Possibly negative symptoms in schizophrenia
- Possibly mixed mood states such as depression or mania with mixed features

Potential Disadvantages

- Expensive

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Symptoms of acute mania/mixed mania

- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms
- Bipolar depression
- Mixtures of manic and depressive symptoms in bipolar disorder

Pearls

- Cariprazine is metabolized into a very long-lasting active metabolite; it is therefore possible that adverse events could appear several weeks after initiation of cariprazine due to accumulation of cariprazine and major metabolites over time
- It is also possible that cariprazine or its very long-lasting active metabolites could be developed as an “oral depot,” namely a very long-lasting oral formulation for weekly or even monthly oral administration
- Based on short-term clinical trials, cariprazine appears to have a favorable metabolic profile, with changes in triglycerides, fasting glucose, and cholesterol similar to placebo; little or no weight gain in bipolar depression, and a small amount of weight gain in bipolar mania and schizophrenia studies especially at higher doses
- Approved as a monotherapy for bipolar depression, with post hoc analyses showing efficacy in bipolar depression with mixed features
- Approved as a monotherapy for acute bipolar mania, with post hoc analyses showing efficacy in bipolar mania with mixed features

- May be one of the best choices for treatment across the spectrum of bipolar disorder, from bipolar depression, to mixed states of bipolar depression and mania, to acute bipolar mania
- Cariprazine is in late-stage testing for augmentation of SSRIs/SNRIs in unipolar major depression
- Approved in Europe for the negative symptoms of schizophrenia
- D3-preferring (over D2) actions represent a novel pharmacologic profile among antipsychotics, especially at lower doses; clinical advantages of this profile remain to be determined but animal models suggest that targeting D3 receptors may have advantages for mood, cognition, negative symptoms, and substance abuse
- All antipsychotics bind to the D3 receptor in vitro, but cariprazine has affinity for the D3 receptor much greater than dopamine itself has for the D3 receptor. Cariprazine is the most potent antipsychotic and one of the only antipsychotics with functional D3 partial agonism in vivo in the living human brain. This means that at therapeutic dosing, cariprazine may have a unique action as a partial agonist at the D3 receptor
- Cariprazine may exert its efficacy across the bipolar spectrum by partial agonist actions at D2 receptors, mostly blocking them in limbic areas, to treat mania and psychosis while having simultaneous partial agonist actions at D3 receptors, especially in the substantia nigra/ventral tegmental area, leading to enhanced dopamine release in prefrontal cortex to treat mood, cognition, and

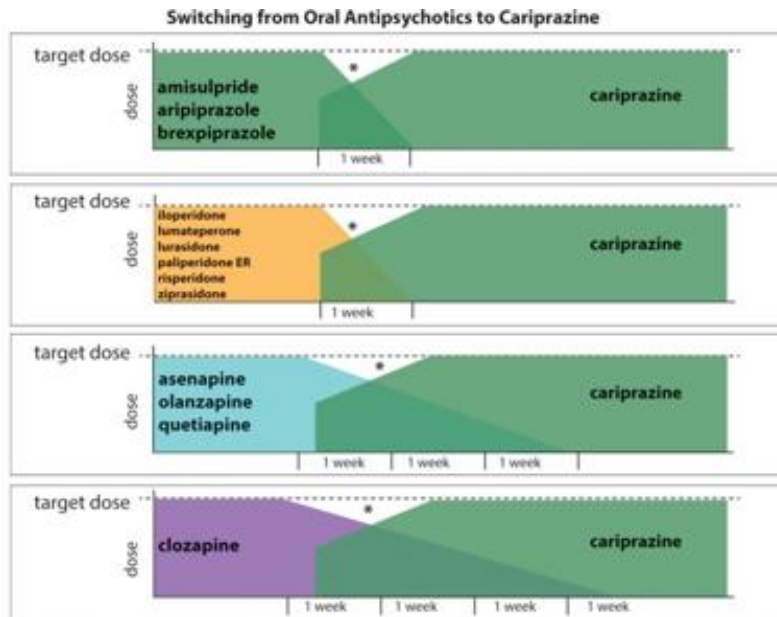
negative symptoms in both schizophrenia and across the mood disorder spectrum

The Art of Switching

Switching from Oral Antipsychotics to Cariprazine

- It is advisable to begin cariprazine at an intermediate dose and build the dose rapidly over 3–7 days
- Clinical experience has shown that asenapine, quetiapine, and olanzapine should be tapered off slowly over a period of 3–4 weeks, to allow patients to readapt to the withdrawal of blocking cholinergic, histaminergic, and alpha 1 receptors
- Clozapine should always be tapered off slowly, over a period of 4 weeks or more
- All antipsychotics bind to the D3 receptor in vitro, but cariprazine has affinity for the D3 receptor much greater than dopamine itself has for the D3 receptor. Cariprazine is the most potent antipsychotic and one of the only antipsychotics with functional D3 partial agonism in vivo in the living human brain. This means that at therapeutic dosing, cariprazine may have a unique action as partial agonist at the D3 receptor

* Benzodiazepine or anticholinergic medication can be administered during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis



Suggested Reading

Choi YK , Adham N , Kiss B , Gyertyn I , Tarazi FI. Long-term effects of cariprazine exposure on dopamine receptor subtypes . *CNS Spectr* 2014 ;**19** (3):268–77 .

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Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Vieta E , Durgam S , Lu K , et al. Effect of cariprazine across the symptoms of mania in bipolar I disorder: analyses of pooled data from phase II/III trials . *Eur Neuropsychopharmacol* 2015 ;**25** (11):1882 –91.

Vieta E , Earley WR , Burgess MV , et al. Longterm safety and tolerability of cariprazine as adjunctive therapy in major depressive disorder. *Int Clin Psychopharmacol* 2019 ;34(2):76–83.

Chlordiazepoxide

Therapeutics

Brands

- Limbitrol
- Librium
- Librax
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (anxiolytic)

Commonly Prescribed for

(bold for FDA approved)

- **Anxiety disorders**
- **Symptoms of anxiety**
- **Preoperative apprehension and anxiety**

- **Withdrawal symptoms of acute alcoholism**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibits neuronal activity presumably in amygdala-centered fear circuits to provide therapeutic benefits in anxiety disorders

How Long Until It Works

- Some immediate relief with first dosing is common; can take several weeks with daily dosing for maximal therapeutic benefit

If It Works

- For short-term symptoms of anxiety –after a few weeks, discontinue use or use on an “as-needed” basis
- For chronic anxiety disorders, the goal of treatment is complete remission of symptoms as well as prevention of future relapses
- For chronic anxiety disorders, treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped

- For long-term symptoms of anxiety, consider switching to an SSRI or SNRI for long-term maintenance
- If long-term maintenance with a benzodiazepine is necessary, continue treatment for 6 months after symptoms resolve, and then taper dose slowly
- If symptoms reemerge, consider treatment with an SSRI or SNRI, or consider restarting the benzodiazepine; sometimes benzodiazepines have to be used in combination with SSRIs or SNRIs for best results

If It Doesn't Work

- Consider switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy, especially cognitive behavioral psychotherapy
- Consider presence of concomitant substance abuse
- Consider presence of clordiazepoxide abuse
- Consider another diagnosis, such as a comorbid medical condition



Best Augmenting Combos for Partial Response or Treatment Resistance

- Benzodiazepines are frequently used as augmenting agents for antipsychotics and mood stabilizers in the treatment of psychotic and bipolar disorders

- Benzodiazepines are frequently used as augmenting agents for SSRIs and SNRIs in the treatment of anxiety disorders
- Not generally rational to combine with other benzodiazepines
- Caution if using as an anxiolytic concomitantly with other sedative hypnotics for sleep

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness

- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Especially at initiation of treatment or when dose increases
- Tolerance often develops over time

What to Do About Side Effects

- Wait

- Wait
- Wait
- Lower the dose
- Take largest dose at bedtime to avoid sedative effects during the day
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Mild to moderate anxiety: 15–40 mg/day in 3–4 doses
- Severe anxiety: 60–100 mg/day in 3–4 doses

Dosage Forms

- Capsule 2.5 mg, 5 mg, 10 mg, 25 mg

How to Dose

- Mild to moderate anxiety: 15–40 mg/day in 3–4 doses
- Severe anxiety: 60–100 mg/day in 3–4 doses

Dosing Tips

Use lowest possible effective dose for the shortest possible period of time (a benzodiazepine-sparing strategy)

Assess need for continued treatment regularly

Risk of dependence may increase with dose and duration of treatment

For interdose symptoms of anxiety, can either increase dose or maintain same total daily dose but divide into more frequent doses

Can also use an as-needed occasional “top-up” dose for interdose anxiety

Because anxiety disorders can require higher doses, the risk of dependence may be greater in these patients

Some severely ill patients may require doses higher than the generally recommended maximum dose

Frequency of dosing in practice is often greater than predicted from half-life, as duration of biological activity is often shorter than pharmacokinetic terminal half-life

Overdose

- Fatalities can occur; hypotension, tiredness, ataxia, confusion, coma

Long-Term Use

- Evidence of efficacy for up to 16 weeks

- Risk of dependence, particularly for treatment periods longer than 12 weeks, and especially in patients with past or current polysubstance abuse

Habit Forming

- Chlordiazepoxide is a Schedule IV drug
- Patients may develop dependence and/or tolerance with long-term use

How to Stop

- Patients with history of seizure may seize upon withdrawal, especially if withdrawal is abrupt
- Taper by 10 mg every 3 days to reduce chances of withdrawal effects
- For difficult-to-taper patients, consider reducing dose much more slowly after reaching 20 mg/day, perhaps by as little as 5 mg per week or less
- For other patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

- Be sure to differentiate reemergence of symptoms requiring reinstitution of treatment from withdrawal symptoms
- Benzodiazepine-dependent anxiety patients and insulin-dependent diabetics are not addicted to their medications. When benzodiazepine-dependent patients stop their medication, disease symptoms can reemerge, disease symptoms can worsen (rebound), and/or withdrawal symptoms can emerge

Pharmacokinetics

- Elimination half-life 24–48 hours



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy

- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Dosage changes should be made in collaboration with prescriber
- Use with caution in patients with pulmonary disease; rare reports of death after initiation of benzodiazepines in patients with severe pulmonary impairment
- History of drug or alcohol abuse often creates greater risk for dependency
- Some depressed patients may experience a worsening of suicidal ideation
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)

Do Not Use

- If patient has angle-closure glaucoma
- If there is a proven allergy to chlordiazepoxide or any benzodiazepine

Special Populations

Renal Impairment

- Initial 10–20 mg/day in 2–4 doses; increase as needed

Hepatic Impairment

- Initial 10–20 mg/day in 2–4 doses; increase as needed

Cardiac Impairment

- Benzodiazepines have been used to treat anxiety associated with acute myocardial infarction

Elderly

- Initial 10–20 mg/day in 2–4 doses; increase as needed
- Elderly patients may be more sensitive to sedative effects



Children and Adolescents

- Not recommended for use in children under age 6
- Initial 10–20 mg/day in 2–4 doses; may increase to 20–30 mg/day in 2–3 doses if ineffective
- Hyperactive children should be monitored for paradoxical effects
- Long-term effects of chlordiazepoxide in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Possible increased risk of birth defects when benzodiazepines are taken during pregnancy
- Because of the potential risks, chlordiazepoxide is not generally recommended as treatment for anxiety during pregnancy, especially during first trimester
- Drug should be tapered if discontinued
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Unknown if chlordiazepoxide is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects of benzodiazepines on nursing infants have been reported and include feeding difficulties, sedation, and weight loss

The Art of Psychopharmacology

Potential Advantages

- Rapid onset of action

Potential Disadvantages

- Euphoria may lead to abuse
- Abuse especially risky in past or present substance abusers

Primary Target Symptoms

- Panic attacks
- Anxiety

Pearls

- Can be a useful adjunct to SSRIs and SNRIs in the treatment of numerous anxiety disorders, but not used as frequently as some other benzodiazepines
- Not effective for treating psychosis as a monotherapy, but can be used as an adjunct to antipsychotics
- Not effective for treating bipolar disorder as a monotherapy, but can be used as an adjunct to mood stabilizers and antipsychotics
- Can both cause depression and treat depression in different patients
- When using to treat insomnia, remember that insomnia may be a symptom of some other primary disorder itself, and thus warrant

evaluation for comorbid psychiatric and/or medical conditions

- * Remains a viable treatment option for alcohol withdrawal
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

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Erstad BL , Cotugno CL. Management of alcohol withdrawal . *Am J Health Syst Pharm* 1995 ;**52** :697 –709 .

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Chlorpromazine

Therapeutics

Brands

- Thorazine
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist, antiemetic)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia (oral)**
- **Severe behavioral problems associated with oppositional defiant disorder or other disruptive behavioral disorders, or for attention deficit hyperactivity disorder (ADHD) in pediatric patients who show excessive motor activity with accompanying**

conduct disorders (oral, intramuscular for acute, severe agitation in hospitalized patients)

- **Acute psychosis (intramuscular)**
- **Nausea, vomiting (oral, rectal, intramuscular, intravenous)**
- **Acute intermittent porphyria (oral, intramuscular)**
- **Tetanus (intramuscular, adjunct)**
- **Intractable hiccups (oral, intramuscular, intravenous)**
- **Bipolar disorder**
- **Restlessness and apprehension before surgery**



How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and improving other behaviors
- Combination of dopamine D2, histamine H1, and cholinergic M1 blockade in the vomiting center may reduce nausea and vomiting

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior
- Immediate and short-term (a few hours) relief of severe behavioral problems if given during an acute exacerbation on an “as needed” prn basis (most common use in children)
- Actions on nausea and vomiting are immediate